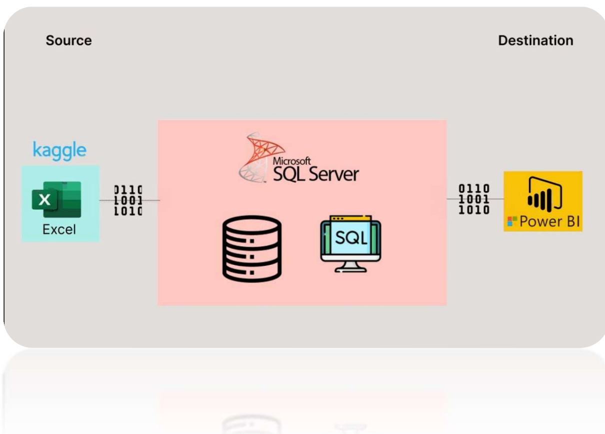

DATA ANALYST WITH POWER BI (INCLUDING EXCEL & SQL)

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ABSTRACT

Data Analyst with expertise in Power BI, Excel, and SQL. Skilled in transforming complex data into actionable insights through interactive dashboards, reports, and data models. Experienced in data cleaning, analysis, and visualization to support business decision-making.

Week 1: Introduction to Data Analytics & Excel Essentials

What Are Data Analytics?

Data Analytics is the process of examining, cleaning, transforming, and interpreting data to discover useful information, draw conclusions, and support decision-making. It involves using tools and techniques to find patterns, trends, correlations, and anomalies in data.

There are four main types of data analytics:

Descriptive Analytics {What happened}?

Summarizes past data to understand what has happened.

e.g., monthly sales reports, website traffic trends

Diagnostic Analytics {Why did it happen}?

Analyzes data to determine why something happened.

e.g., investigating a sales drop in a specific region

Predictive Analytics {What is likely to happen}?

Uses data and models to forecast what is likely to happen.

e.g., forecasting customer churn

Prescriptive Analytics {What should be done}?

Suggests actions to take for optimal outcomes based on data insights.

(e.g., recommending actions to improve performance)

Role of a Data Analyst

A Data Analyst plays a crucial role in helping organizations make data-driven decisions. Here's what they typically do:

Key Responsibilities:

1. **Data Collection & Extraction**

   Gather data from various sources like databases (SQL-Structured Query Language), spreadsheets (Excel), APIs (Application Programming Interface), and software tools.

2. **Data Cleaning & Preparation**

   Remove errors, handle missing values, and format data to ensure its accurate and ready for analysis.

3. **Data Analysis**

   Use statistical methods and tools to identify trends, correlations, and patterns.

4. **Data Visualization**

   Create dashboards and visual reports using tools like **Power BI**, Excel, or Tableau to make insights easy to understand.

5. **Reporting**

   Present findings to stakeholders and decision-makers clearly and effectively.

6. **Automation**

   Build automated reports and workflows to save time and reduce manual tasks.

7. **Collaboration**

   Work with business teams, IT, and management to understand data needs and provide actionable insights.

Overview of Excel, Power Bi and SQL in Data Analysis

Excel – Spreadsheet-Based Analysis Tool

Purpose: Ideal for quick, manual data analysis and visualization.

Strengths:

- User-friendly interface.
- Built-in functions for stats, pivot tables, charts.
- Good for small datasets and ad-hoc analysis.

Common Uses: Data cleaning, trend analysis, dashboards, financial modeling.

SQL (Structured Query Language) – Data Extraction Tool

Purpose: Used to query, manipulate, and manage data stored in relational databases.

Strengths:

- Efficiently handles large datasets.

- Enables filtering, joining, and aggregating data from multiple tables.
- Essential for backend data access and reporting.

Common Uses: Extracting specific data, preparing datasets for analysis, automating data retrieval.

Power BI – Business Intelligence & Visualization Tool

Purpose: Turns raw data into interactive dashboards and reports.

Strengths:

- Advanced visualizations and data modeling.
- Connects to multiple data sources (Excel, SQL, APIs, etc.).
- Supports real-time dashboards and sharing across teams.

Common Uses: Data storytelling, performance tracking, executive reporting, real-time analytics.

Summary Table:

Tool	Key Use	Best For
Excel	Data manipulation	Small to mid-size datasets
SQL	Data extraction	Querying databases efficiently
Power BI	Data visualization & reporting	Creating dynamic dashboards & insights

Data types, Formatting, Filtering, Sorting

Data Types

Defines the kind of data stored in a cell, column, or field.

Data Type	Description	Examples
Text/String	Alphanumeric characters	"Product A", "John"
Number	Numeric values (integers or decimals)	100, 45.67
Date/Time	Calendar dates and time values	16-Aug-2025, 10:30 AM
Boolean	True/False or Yes/No values	TRUE, FALSE

Formatting

Improving the visual presentation of data without changing its actual value.

Learn using Practical

Filtering

Showing only the rows that meet specific criteria.

- **Example:** Filter a sales table to show only entries where "Region = North" or "Sales > $10,000".
- Used for focused analysis and removing irrelevant data temporarily.

Learn using Practical

Sorting

Arranging data in a particular order.

- **Ascending**: A–Z or smallest to largest (e.g., sort names or prices)
- **Descending**: Z–A or largest to smallest
- Useful for ranking, comparisons, and identifying top/bottom values.

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Using formulas and functions:

Learn using Practical, Learn different formulas

Basic charts and graphs in Excel:

Chart Type	Purpose	Best For
Column Chart	Compares values vertically using columns	Sales by product, monthly revenue
Bar Chart	Compares values horizontally	Ranking items, category comparisons
Line Chart	Shows trends over time	Stock prices, monthly sales, growth
Pie Chart	Shows percentage contribution of parts to a whole	Market share, budget distribution

Chart Type	Purpose	Best For
Doughnut Chart	Similar to pie chart, but with a blank center	Showing part-to-whole relationships
Area Chart	Like a line chart, but filled with color	Cumulative values over time
Scatter Plot	Shows relationships or correlation between two variables	Height vs. weight, sales vs. ads
Combo Chart	Combines two chart types (e.g., column + line)	Comparing two datasets with different scales
Histogram	Shows distribution of numerical data	Exam scores, age groups
PivotChart	Dynamic chart linked to a PivotTable	Interactive dashboards

Exercise:

Excel Practice Exercise -------- Sales Data Analysis

Scenario:

You're analyzing monthly sales performance for 5 salespeople. You need to:

- Calculate total sales and average sales
- Count how many met the target
- Flag performance with IF
- Visualize results using basic charts likely Column Chart, Pie Chart

Create This Table in Excel

Name	Region	Sales	Target
John	East	12000	10000
Priya	West	9000	10000
Ahmed	North	15000	10000
Sara	South	8000	10000
David	East	11000	10000